

hello guys !  
I am Muskan Garg



# HOSPITAL MANAGEMENT DATA ANALYSIS THROUGH SQL





**Q1) CREATE A TABLE SHOWING THE TOTAL NUMBER OF PATIENTS, DOCTORS , APPOINTMENTS, TREATMENTS AND TOTAL REVENUE GENERATED BY THE HOSPITAL.**



```
SELECT (SELECT COUNT(patient_id) FROM patients_data) AS patient_count,  
(SELECT COUNT(appointment_id) FROM appointments_record) As appointment_count,  
(SELECT COUNT(doctor_id) FROM doctors_record) AS doctor_count,  
(SELECT COUNT(treatment_id) FROM treatment_record) AS treatment_count,  
(SELECT SUM(amount) FROM billing_record) AS total_revenue;
```

|   | patient_count<br>bigint | appointment_count<br>bigint | doctor_count<br>bigint | treatment_count<br>bigint | total_revenue<br>numeric |
|---|-------------------------|-----------------------------|------------------------|---------------------------|--------------------------|
| 1 | 50                      | 200                         | 10                     | 200                       | 551249.85                |



Q2) ANALYZE THE MONTHLY REVENUE TREND AND IDENTIFY THE MONTH WITH THE HIGHEST AND LOWEST REVENUE.



```
SELECT MIN(monthly_revenue) AS lowest_revenue,MAX(monthly_revenue) AS highest_revenue
FROM(SELECT EXTRACT(Month FROM bill_date) AS bill_month ,
SUM(amount) AS monthly_revenue
FROM billing_record
GROUP BY bill_month) AS revenue_details;
```



|   | lowest_revenue<br>numeric | highest_revenue<br>numeric |
|---|---------------------------|----------------------------|
| 1 | 27569.71                  | 64271.54                   |

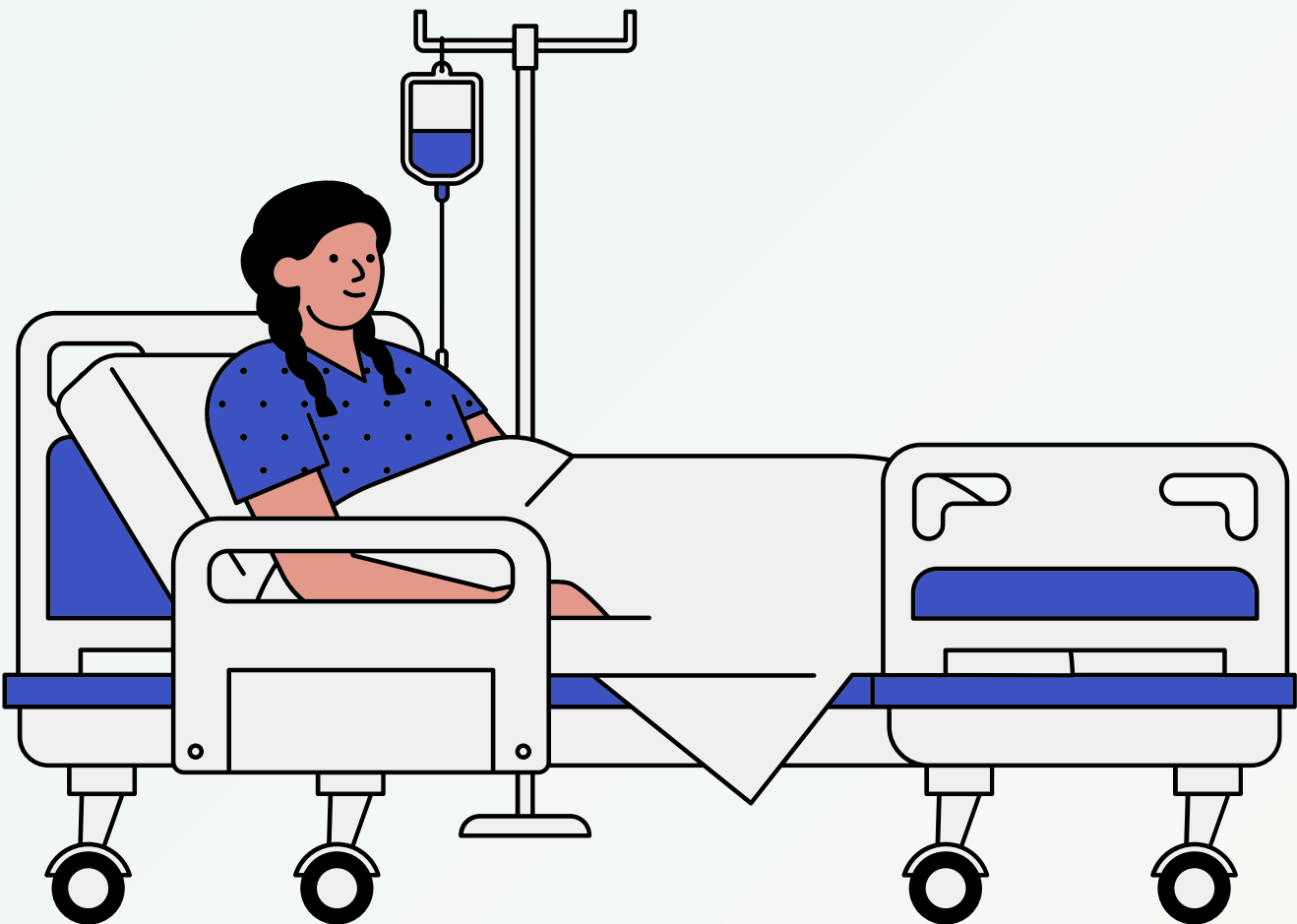




Q3) IDENTIFY THE TOP 10 PATIENTS BASED ON THEIR TOTAL SPENDING ON TREATMENTS AND BILLING.



```
SELECT p.patient_id,p.first_name,p.last_name,(b.amount+t.cost) AS total_amount_spend
FROM patients_data p
INNER JOIN
billing_record b
on p.patient_id=b.patient_id
inner join
treatment_record t
on b.treatment_id=t.treatment_id
order by total_amount_spend desc limit 10;
```



|    | patient_id<br>[PK] character varying (500) | first_name<br>character varying (500) | last_name<br>character varying (500) | total_amount_spend<br>numeric |
|----|--------------------------------------------|---------------------------------------|--------------------------------------|-------------------------------|
| 1  | P024                                       | Sarah                                 | Brown                                | 9947.26                       |
| 2  | P017                                       | Jane                                  | Jones                                | 9932.36                       |
| 3  | P021                                       | Michael                               | Wilson                               | 9929.42                       |
| 4  | P050                                       | Laura                                 | Wilson                               | 9921.30                       |
| 5  | P044                                       | Robert                                | Taylor                               | 9890.06                       |
| 6  | P047                                       | Jane                                  | Moore                                | 9780.50                       |
| 7  | P038                                       | David                                 | Smith                                | 9692.40                       |
| 8  | P029                                       | David                                 | Smith                                | 9668.04                       |
| 9  | P033                                       | Michael                               | Wilson                               | 9666.34                       |
| 10 | P049                                       | David                                 | Moore                                | 9618.62                       |

## Q4) FIND THE TOP 5 DOCTORS WHO GENERATED THE HIGHEST REVENUE FOR THE HOSPITAL.

```
SELECT d.doctor_id,d.first_name,d.last_name,SUM(b.amount) AS revenue
FROM doctors_record d
INNER JOIN
appointments_record a
ON d.doctor_id=a.doctor_id
inner join
billing_record b
ON a.patient_id=b.patient_id
GROUP BY d.doctor_id
ORDER BY revenue DESC LIMIT 5;
```



|   | doctor_id<br>[PK] character varying (500)  | first_name<br>character varying (500)  | last_name<br>character varying (500)  | revenue<br>numeric  |
|---|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| 1 | D005                                                                                                                            | Sarah                                                                                                                       | Taylor                                                                                                                     | 394356.89                                                                                                |
| 2 | D006                                                                                                                            | Alex                                                                                                                        | Davis                                                                                                                      | 339280.84                                                                                                |
| 3 | D001                                                                                                                            | David                                                                                                                       | Taylor                                                                                                                     | 336851.92                                                                                                |
| 4 | D010                                                                                                                            | Linda                                                                                                                       | Wilson                                                                                                                     | 304846.08                                                                                                |
| 5 | D003                                                                                                                            | Jane                                                                                                                        | Smith                                                                                                                      | 279957.14                                                                                                |



**Q5) COMPARE THE NUMBER OF APPOINTMENTS HANDLED BY EACH DOCTOR AND RANK THEM FROM HIGHEST TO LOWEST.**



```
SELECT d.doctor_id,d.first_name,d.last_name,
COUNT(a.appointment_id) AS count_appointments,
RANK() OVER(ORDER BY COUNT(a.appointment_id)) AS doctor_rank
FROM doctors_record d
INNER JOIN
appointments_record a
ON d.doctor_id=a.doctor_id
GROUP BY d.doctor_id
ORDER BY doctor_rank DESC;
```



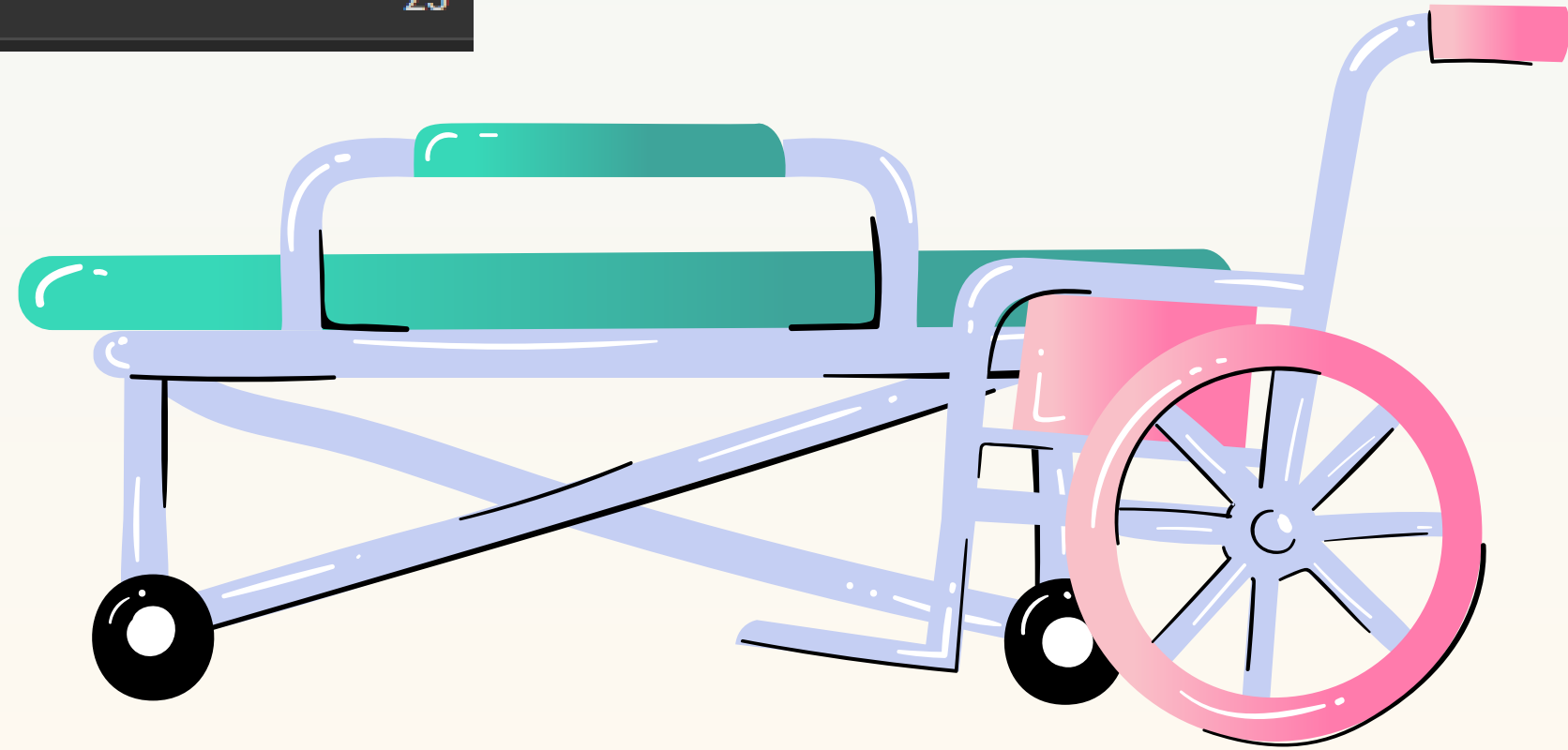
|    | doctor_id<br>[PK] character varying (500) | first_name<br>character varying (500) | last_name<br>character varying (500) | count_appointments | doctor_rank |
|----|-------------------------------------------|---------------------------------------|--------------------------------------|--------------------|-------------|
| 1  | D005                                      | Sarah                                 | Taylor                               | 29                 | 10          |
| 2  | D001                                      | David                                 | Taylor                               | 25                 | 9           |
| 3  | D006                                      | Alex                                  | Davis                                | 24                 | 8           |
| 4  | D003                                      | Jane                                  | Smith                                | 22                 | 7           |
| 5  | D002                                      | Jane                                  | Davis                                | 21                 | 6           |
| 6  | D010                                      | Linda                                 | Wilson                               | 19                 | 5           |
| 7  | D009                                      | Sarah                                 | Smith                                | 17                 | 4           |
| 8  | D008                                      | Linda                                 | Brown                                | 16                 | 3           |
| 9  | D004                                      | David                                 | Jones                                | 14                 | 2           |
| 10 | D007                                      | Robert                                | Davis                                | 13                 | 1           |

**Q6) DETERMINE WHICH DOCTOR HAS TREATED THE HIGHEST NO. OF UNIQUE PATIENTS.**



```
SELECT d.doctor_id,d.first_name,d.last_name,
COUNT(DISTINCT a.patient_id) AS count_unique_patient
FROM doctors_record d
INNER JOIN
appointments_record a
ON d.doctor_id=a.doctor_id
GROUP BY d.doctor_id
ORDER BY count_unique_patient DESC LIMIT 1;
```

|   | doctor_id<br>[PK] character varying (500) | first_name<br>character varying (500) | last_name<br>character varying (500) | count_unique_patient<br>bigint |
|---|-------------------------------------------|---------------------------------------|--------------------------------------|--------------------------------|
| 1 | D005                                      | Sarah                                 | Taylor                               | 23                             |





**Q7) IDENTIFY THE MOST FREQUENTLY PERFORMED TREATMENT AND CALCULATE THE REVENUE GENERATED FROM EACH TREATMENT TYPE.**

```
SELECT treatment_type,COUNT(treatment_type) AS treatment_frequency,  
SUM(cost) AS treatment_cost FROM treatment_record  
GROUP BY treatment_type  
ORDER BY treatment_frequency DESC;
```



|   | treatment_type<br>character varying (500) | treatment_frequency<br>bigint | treatment_cost<br>numeric |
|---|-------------------------------------------|-------------------------------|---------------------------|
| 1 | Chemotherapy                              | 49                            | 128855.68                 |
| 2 | X-Ray                                     | 41                            | 110653.67                 |
| 3 | ECG                                       | 38                            | 96224.24                  |
| 4 | MRI                                       | 36                            | 116098.16                 |
| 5 | Physiotherapy                             | 36                            | 99418.10                  |



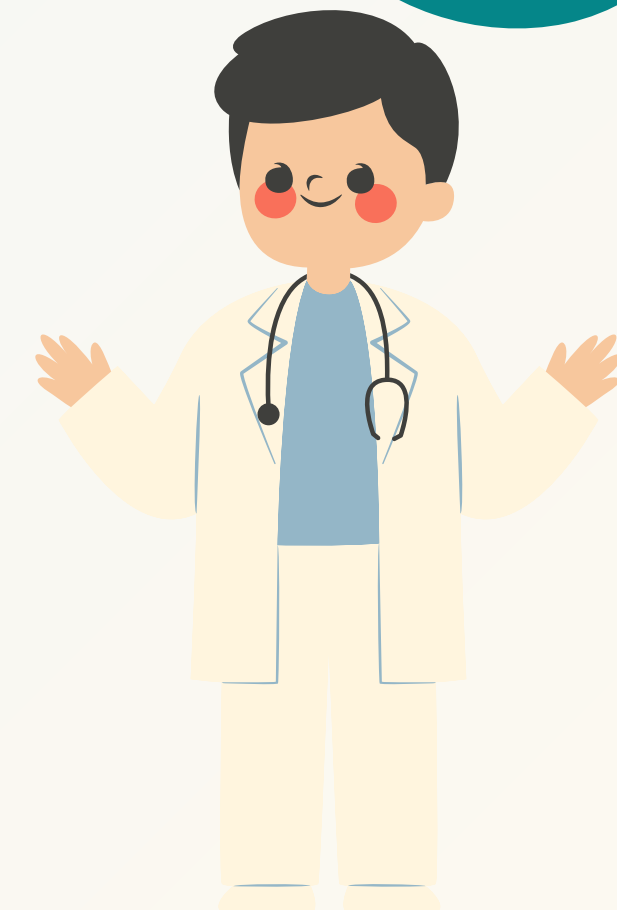


## Q8) ANALYZE APPOINTMENT STATUS BY CALCULATING THE PERCENTAGE OF COMPLETED, CANCELLED AND PENDING APPOINTMENTS.



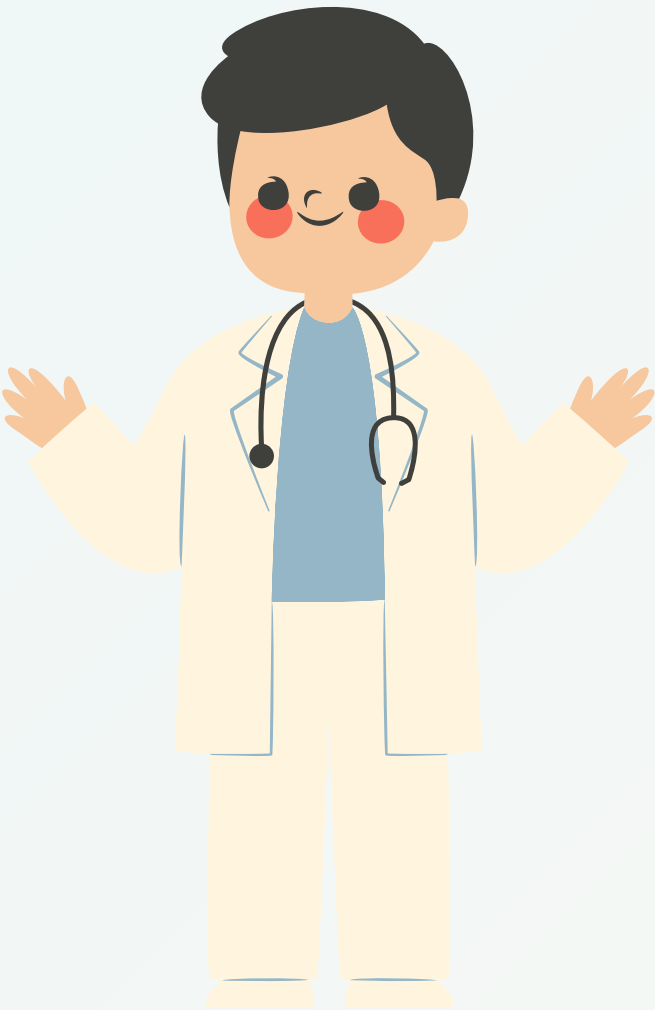
```
SELECT status,COUNT(status)*100/SUM(COUNT(status)) OVER()  
AS appointment_status_percentage  
FROM appointments_record  
GROUP BY status;
```

|   | status<br>character varying (500)  | appointment_status_percentage<br>numeric  |
|---|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| 1 | Scheduled                                                                                                               | 25.500000000000000000                                                                                                          |
| 2 | No-show                                                                                                                 | 26.000000000000000000                                                                                                          |
| 3 | Completed                                                                                                               | 23.000000000000000000                                                                                                          |
| 4 | Cancelled                                                                                                               | 25.500000000000000000                                                                                                          |

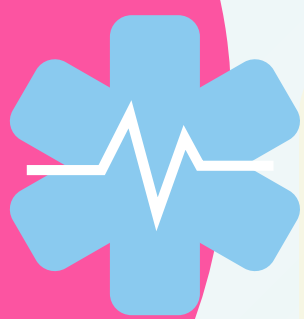


Q9) FIND THE BUSIEST DAY AND BUSIEST MONTH BASED ON THE NUMBER OF APPOINTMENTS SCHEDULED.

```
SELECT appointment_date,EXTRACT(MONTH FROM appointment_date) AS month,
EXTRACT(day FROM appointment_date) AS day,
COUNT(appointment_id) AS total_appointments
FROM appointments_record
GROUP BY appointment_date
ORDER BY total_appointments DESC LIMIT 1;
```



|   | appointment_date  | month  | day  | total_appointments  |
|---|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
|   | date                                                                                                   | numeric                                                                                     | numeric                                                                                   | bigint                                                                                                   |
| 1 | 2023-08-16                                                                                             | 8                                                                                           | 16                                                                                        | 5                                                                                                        |



**Q10) CREATE A SPECIALIZATION WISE TABLE SHOWING THE TOTAL NUMBER OF DOCTORS, APPOINTMENTS, PATIENTS TREATED AND Revenue GENERATED.**



```
SELECT d.specialization,COUNT(t.treatment_id) AS treatment_count,  
SUM(b.amount) AS revenue_generated,COUNT(DISTINCT b.patient_id) AS treated_patient_count,  
COUNT(a.appointment_id) AS appointment_count,  
COUNT(d.doctor_id) AS doctor_count FROM doctors_record d  
inner join appointments_record a  
on d.doctor_id=a.doctor_id  
INNER JOIN billing_record b  
ON b.patient_id=a.patient_id  
INNER JOIN treatment_record t  
ON t.treatment_id=b.treatment_id  
group by d.specialization;
```

|   | specialization<br>character varying (500) 🔒 | treatment_count<br>bigint 🔒 | revenue_generated<br>numeric 🔒 | treated_patient_count<br>bigint 🔒 | appointment_count<br>bigint 🔒 | doctor_count<br>bigint 🔒 |
|---|---------------------------------------------|-----------------------------|--------------------------------|-----------------------------------|-------------------------------|--------------------------|
| 1 | Dermatology                                 | 353                         | 965566.84                      | 37                                | 353                           | 353                      |
| 2 | Oncology                                    | 177                         | 493428.71                      | 22                                | 177                           | 177                      |
| 3 | Pediatrics                                  | 472                         | 1289294.61                     | 47                                | 472                           | 472                      |



Q11) ANALYZE PAYMENT METHODS BY FINDING THE TOTAL Revenue AND PERCENTAGE CONTRIBUTION OF EACH PAYMENT METHOD (CASH, UPI, CARD etc.)



```
SELECT payment_method,SUM(amount)*100/SUM(SUM(amount)) OVER() AS percentage_contribution,  
SUM(amount) AS revenue_generated FROM billing_record  
GROUP BY payment_method;
```



|   | payment_method<br>character varying (500) | percentage_contribution<br>numeric | revenue_generated<br>numeric |
|---|-------------------------------------------|------------------------------------|------------------------------|
| 1 | Insurance                                 | 33.0449577446597038                | 182160.28                    |
| 2 | Credit Card                               | 36.5319700313750652                | 201382.43                    |
| 3 | Cash                                      | 30.4230722239652310                | 167707.14                    |



**Q12) RANK DOCTORS WITH EACH SPECIALIZATION BASED ON THE REVENUE THEY GENERATED USING WINDOW FUNCTIONS.**



```
SELECT d.specialization,sum(b.amount) AS revenue_generated,  
RANK() OVER(order by sum(b.amount) DESC ) AS patient_rank  
FROM doctors_record d inner join appointments_record a  
on d.doctor_id=a.doctor_id  
inner join billing_record b  
on a.patient_id=b.patient_id  
group by d.specialization;
```

|   | specialization<br>character varying (500)  | revenue_generated<br>numeric  | patient_rank<br>bigint  |
|---|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 1 | Pediatrics                                                                                                                      | 1289294.61                                                                                                         | 1                                                                                                            |
| 2 | Dermatology                                                                                                                     | 965566.84                                                                                                          | 2                                                                                                            |
| 3 | Oncology                                                                                                                        | 493428.71                                                                                                          | 3                                                                                                            |



Q13) CREATE A TABLE SHOWING TOTAL NUMBER BILLS, TOTAL Revenue AND AVERAGE BILL AMOUNT.



```
SELECT EXTRACT(MONTH FROM bill_date) AS payment_month,
SUM(amount) AS total_revenue,AVG(amount) AS average_revenue,COUNT(bill_id) AS total_bills
FROM billing_record
GROUP BY payment_month;
```



|    | payment_month<br>numeric | total_revenue<br>numeric | average_revenue<br>numeric | total_bills<br>bigint |
|----|--------------------------|--------------------------|----------------------------|-----------------------|
| 1  | 3                        | 47304.29                 | 2489.6994736842105263      | 19                    |
| 2  | 1                        | 58701.23                 | 2935.0615000000000000      | 20                    |
| 3  | 4                        | 64271.54                 | 2570.8616000000000000      | 25                    |
| 4  | 9                        | 33426.53                 | 3038.7754545454545455      | 11                    |
| 5  | 6                        | 56887.82                 | 3160.4344444444444444      | 18                    |
| 6  | 10                       | 43314.15                 | 3093.8678571428571429      | 14                    |
| 7  | 8                        | 41958.67                 | 2797.2446666666666667      | 15                    |
| 8  | 7                        | 39880.19                 | 2492.5118750000000000      | 16                    |
| 9  | 2                        | 36669.69                 | 2619.2635714285714286      | 14                    |
| 10 | 11                       | 52474.98                 | 3086.7635294117647059      | 17                    |
| 11 | 12                       | 27569.71                 | 2297.4758333333333333      | 12                    |

THANK YOU

